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DATA DESCRIPTOR

In vivo submillimeter diffusion MRI dataset of 9 macaque brains curated for tractography

Alex Valcourt Caron^{1,6}, Nadège Corbin^{2,5,6}, Adrien Boissenin³, Ankur Gupta³,
Tho-Hai Nguyen³, Serge Anandra⁴, Sylvain Miraux^{2,4,5}, Maxime Descoteaux^{1,5},
Laurent Petit^{4,5,7} ✉ & Fabien B. Wagner^{3,7} ✉

In vivo diffusion magnetic resonance imaging (MRI) is critical to access detailed information about the brain microstructure and connectivity. In non-human primates (NHPs), especially macaque monkeys, such acquisitions should be carried out at a submillimeter resolution to enable accurate tractography, i.e. the 3D reconstruction of white matter fiber tracts. However, the lack of standardized methods and openly available datasets suitable for tractography applications has hindered the full deployment of this technique for *in vivo* applications in macaques. Here, we present a new diffusion MRI dataset acquired in 9 macaque monkeys (*Macaca fascicularis*) with clinical technology (3T MRI scanner and commercially available MRI coils) at an isotropic resolution of 0.7 mm, together with T1- and T2-weighted anatomical images at an isotropic resolution of 0.5 mm. This new dataset, called MADI@700 (Macaque Anatomical and Diffusion Imaging at 700 μ m), was curated for tractography applications and includes a diffusion MRI template of the macaque brain. We illustrate the quality of this dataset by the extraction of several representative white matter fiber tracts.

Background & Summary

Since its inception in the mid-80s, diffusion MRI has become the main non-invasive neuroimaging modality to investigate *in vivo* the microstructural architecture and connectivity of the brain, thanks to its sensitivity to the diffusion of water molecules within biological tissues¹. A plethora of innovations in the acquisition sequences and data processing pipelines has led to tremendous progress in our understanding of the structural and functional architecture of the brain, shedding light on its evolution, interindividual variability, and disconnections in neurological disorders and injuries². Such understanding requires the development of diffusion MRI protocols and datasets across multiple species, including NHPs, due to their phylogenetic proximity, strong brain homology, and similar physiology compared to humans. In particular, high-quality diffusion MRI of the macaque brain is crucial for many translational neuroscience studies, such as the development of novel neurotechnologies that aim at modulating specific brain networks³. Although diffusion MRI may be performed at excellent spatial resolutions *ex vivo*, the ability to perform such acquisitions *in vivo* is critical for longitudinal studies where diffusion MRI data are acquired at several time points⁴, or when those images are used to guide subsequent interventions, as in personalized connectomic deep brain stimulation⁵. Additionally, performing diffusion MRI acquisitions *in vivo* avoids the structural alterations that occur in post mortem tissues⁶. Yet, the existence of a dataset of reference for high-resolution *in vivo* diffusion MRI of the macaque brain is still lacking.

The ability of diffusion MRI to accurately characterize the microstructure and connectivity of the brain greatly depends on the spatial resolution of the acquisitions, which is typically limited to 1.25–3 mm in each dimension, with MRI protocols classically used in humans. Only a few studies have achieved an isotropic resolution of the *in vivo* human brain below 1 mm using specialized MRI pulse sequences on clinical 3T scanners^{7,8} or high-gradient-strength Connectome scanners^{9–13}. In NHPs, the resolution of most *in vivo* diffusion MRI studies

¹Sherbrooke Connectivity Imaging Laboratory, Computer Science Department, Université de Sherbrooke, Sherbrooke, QC, Canada. ²Univ. Bordeaux, CNRS, CRMSB, UMR 5536, F-33000, Bordeaux, France. ³Univ. Bordeaux, CNRS, IMN, UMR 5293, F-33000, Bordeaux, France. ⁴Biomedical Imaging platform pIBIO, UAR3767, Univ. Bordeaux, CNRS, Bordeaux, France. ⁵IRP OpTeam, CNRS Biologie, France - Université de Sherbrooke, Canada. ⁶These authors contributed equally: Alex Valcourt Caron, Nadège Corbin. ⁷These authors jointly supervised this work: Laurent Petit, Fabien B Wagner. ✉e-mail: laurent.petit@u-bordeaux.fr; fabien.wagner@u-bordeaux.fr

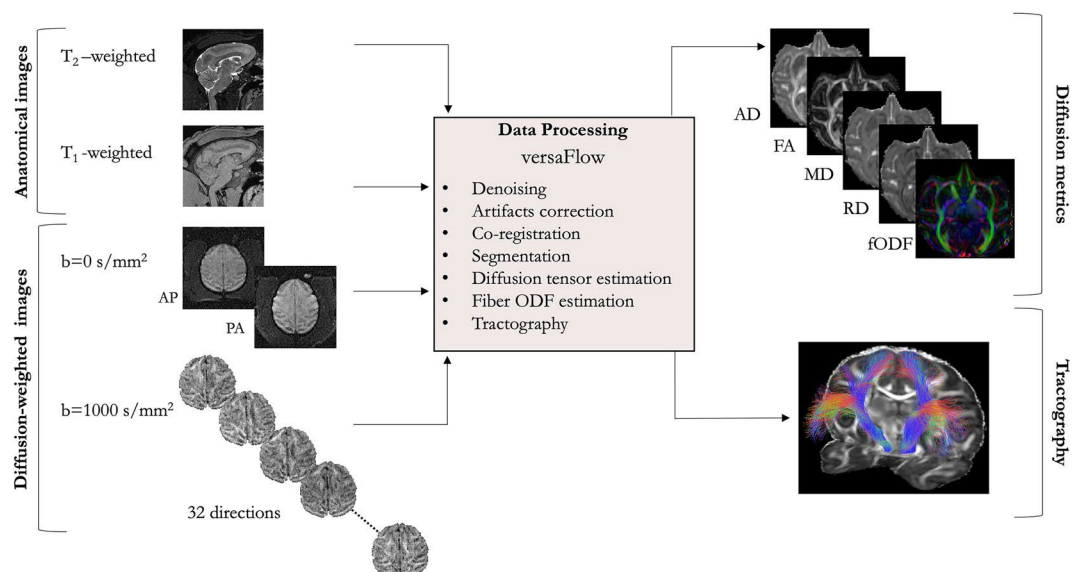


Fig. 1 Experimental workflow used to create the MADI@700 dataset. Diffusion-weighted and anatomical images were acquired *in vivo* for each animal. Data were processed with the *versaFlow* pipeline, resulting in multiple diffusion metrics maps and tractograms.

also remains on the order of 1–2 mm in each dimension^{14–24}. However, the smaller size of the NHP brain requires acquisitions at a submillimeter scale to achieve sufficient accuracy for translational perspectives. Moreover, the existing datasets contain a large degree of variability, as shown in a recent analysis of the PRIME-DE diffusion database^{21,24} using a new multi-resolution pipeline called *versaFlow*²⁵.

To circumvent these limitations, ultra-high field MRI scanners (>9 T) can be employed to image the living brain of small NHP species such as squirrel monkeys²⁶ or marmosets²⁷ and even larger macaque monkeys²⁸, but are not readily available except in specialized facilities. To allow the widespread use and standardization of *in vivo* diffusion MRI studies in macaque monkeys across research centers, submillimeter acquisitions with clinical 3T MRI scanners are therefore highly desirable. Yet, the decrease in signal-to-noise ratio (SNR) associated with the lower magnetic field makes high spatial resolution imaging more challenging. Some groups have developed custom-built MRI coils, either implanted onto the skull²⁹ or external to the animal^{30,31}, achieving isotropic resolutions of 0.7 and 0.9 mm, respectively. Others have replaced the conventional 2D MRI pulse sequences, such as 2D Echo Planar Imaging (EPI), with 3D EPI pulse sequences able to achieve submillimeter resolution while maintaining a reasonable SNR³². As of today, 3D EPI sequences have enabled *in vivo* diffusion MRI at 3T with an isotropic resolution of 0.85 mm in humans⁸, 0.5 mm in a first study in 4 macaques monkeys³³, and 0.4 mm in a subsequent study in 3 macaques when triggered on cardiovascular pulsations³⁴.

Despite these proof-of-concept studies, there is, however, no submillimeter diffusion MRI dataset of the macaque brain at 3T openly available to the community. Here, we provide MADI@700 (Figs. 1, 2), a comprehensive diffusion MRI dataset of 9 macaque brains *in vivo* (5 males and 4 females, 3–23 years old, *Macaca fascicularis*) acquired at an isotropic resolution of 0.7 mm (without any interpolation) using a 3D EPI sequence that we implemented in-house. We further completed this dataset with high-resolution T1-weighted and T2-weighted anatomical images at an isotropic resolution of 0.5 mm, registered to the diffusion MRI images (Fig. 3) and to several anatomical macaque brain atlases (INIA19³⁵, D99³⁶, SARM³⁷, CHARM³⁸). We also provide an extensive quantification of MADI@700 using diffusion metrics derived from our recent pipeline *versaFlow*²⁵, including the fractional anisotropy (FA), mean, axial, and radial diffusivities, the apparent fiber density (AFD) maps, and the white matter fiber orientation distribution function (fODF) (Fig. 4). We curated this dataset for future tractography applications, as illustrated with representative fiber bundles (Figs. 5, 6). Finally, we generated a diffusion MRI template of the macaque brain from our 9 animals as a reference for the community and for future examinations of the interindividual variability of various diffusion metrics.

We believe that the MADI@700 dataset can establish a new standard for *in vivo* diffusion MRI acquisitions of the macaque brain using clinical technology (*i.e.*, a 3T MRI scanner and commercially available MRI coils). To ensure that our methods can be easily replicated, our custom acquisition pulse sequence can be freely downloaded from the Siemens Teamplay C2P exchange platform (DW-3D-SE-EPI sequence) and runs on a 3T Siemens MRI scanner. The MADI@700 dataset has been uploaded to the EBRAINS infrastructure (<https://doi.org/10.25493/XGXR-BRD>)³⁹, as well as the PRIME-DE database, a major platform for the exchange of datasets and protocols for NHP neuroimaging (https://fcon_1000.projects.nitrc.org/indi/PRIME/MADI@700.html). We hope that these measures will allow to further enrich it in the future by the addition of new contributions from our group and others.

Methods

Animals. We acquired data from 5 male and 4 female healthy macaque monkeys (*Macaca fascicularis*) aged between 3 and 23 years old at the time of experiments. Information on sex, age, weight, and brain volume is provided in Table 1. Experiments were approved by the Institutional Animal Care and Use Committee

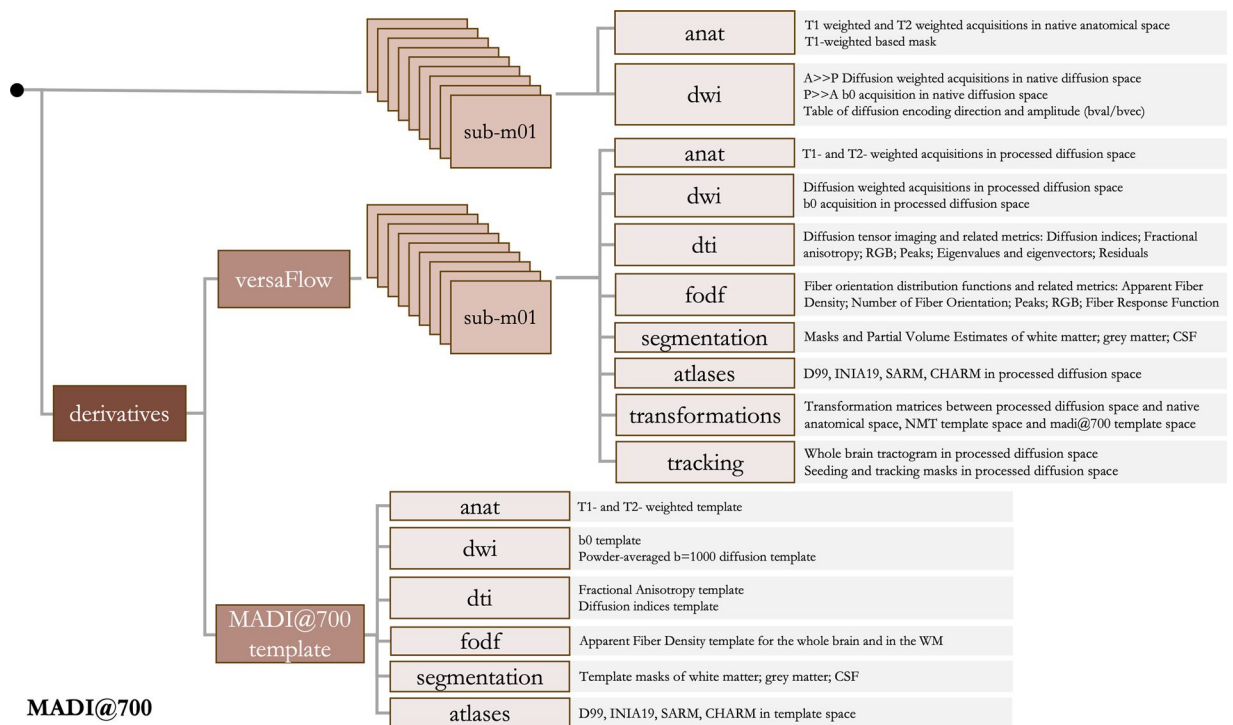


Fig. 2 Data structure of the MADI@700 dataset. Raw data and processed maps are available on the EBRAINS and PRIME-DE platforms for all animals. The anatomical and diffusion template built from all animals is also provided.

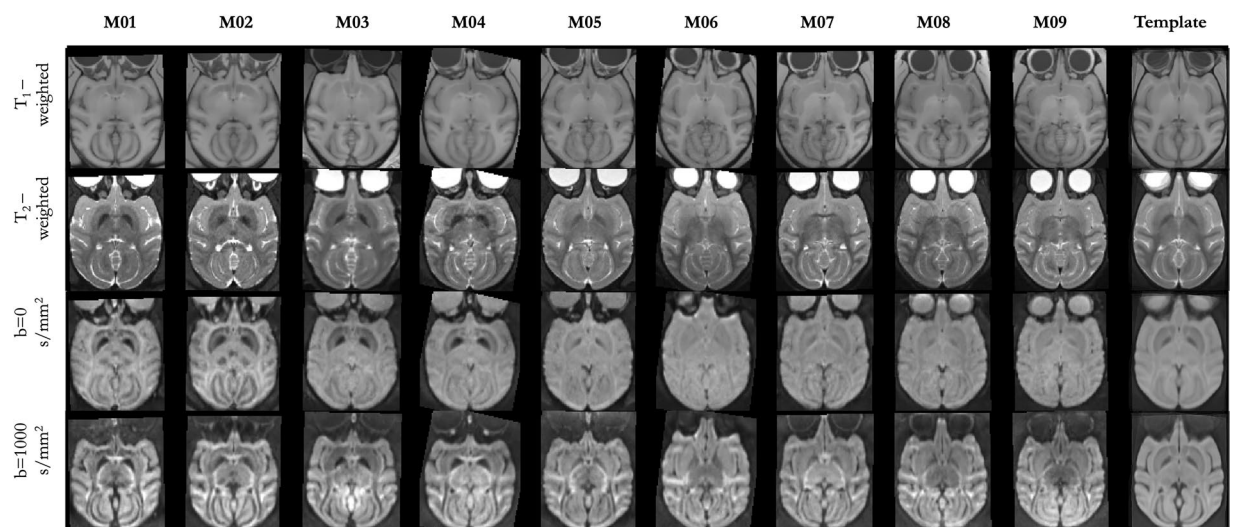


Fig. 3 Raw anatomical and diffusion images from all 9 subjects and the resulting template. Note that the image in the last row and the last column corresponds to the powder-averaged $b = 1000 \text{ s/mm}^2$ shell of the diffusion weighted images in the template space.

of Bordeaux (CEEA-50, France) and the French Ministry of Higher Education and Research (project numbers APAFIS#26048-2020061216228969-v5 and APAFIS#25587-2020052908471774-v4) in accordance with the European Union directive of 22 September 2010 (2010/63/EU) on the protection of animals used for scientific purposes. Animals were housed in groups whenever possible, in enclosures designed according to the European directive 2010/63/EU in the non-human primate facility of the Institute of Neurodegenerative Diseases (CNRS UMR5293, University of Bordeaux, animal experimentation registration number D33063268). Environmental enrichment included background music, foraging toys and bedding, as well as cognitive games performed on a touchscreen system intermittently accessible from their housing enclosure.

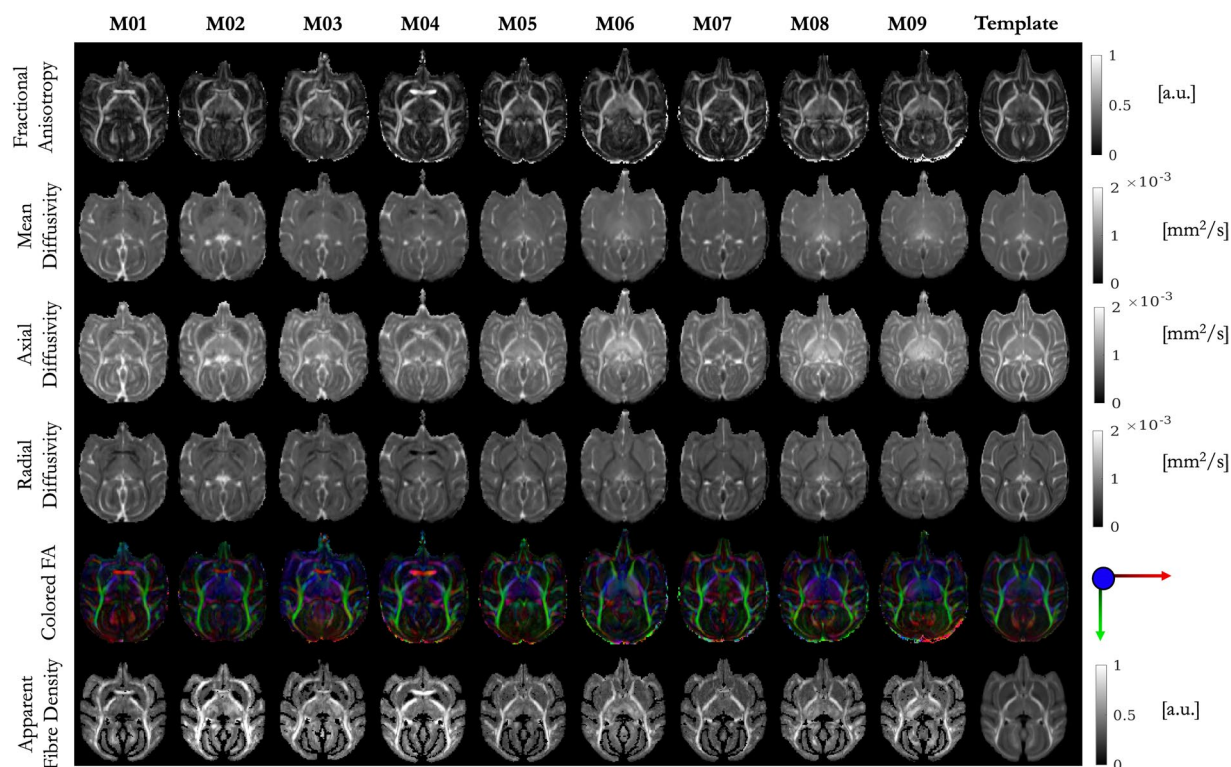


Fig. 4 Diffusion metrics computed by *versaFlow* for all 9 subjects and the resulting template. The output images result from the *versaFlow* processing applied to each of the 9 animals. The template combining data from all 9 animals is shown in the last column.

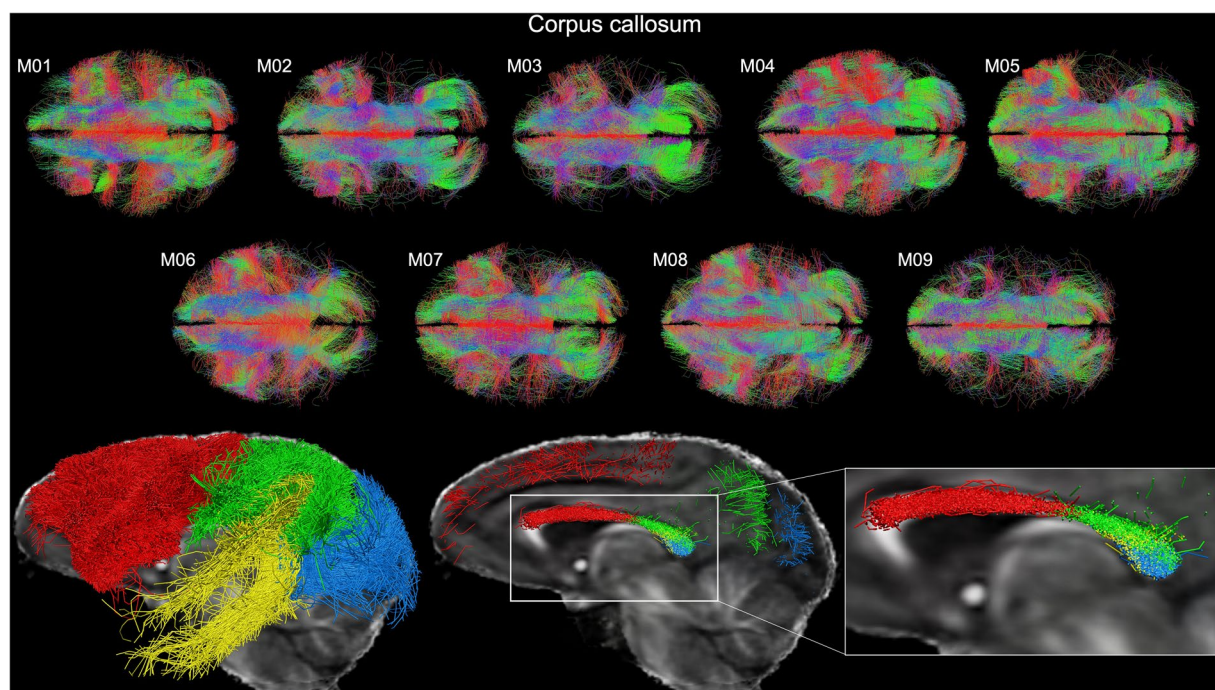


Fig. 5 Corpus callosum extracted from whole-brain tractograms for individual subjects (top) and combined across them (bottom).

MRI data acquisition. *Equipment and animal preparation.* MRI data acquisition was carried out at the Biomedical Imaging Platform of the University of Bordeaux (CNRS UAR3767, University of Bordeaux, animal experimentation registration number A33-063-930). All experiments were performed with a 3T Prisma Siemens

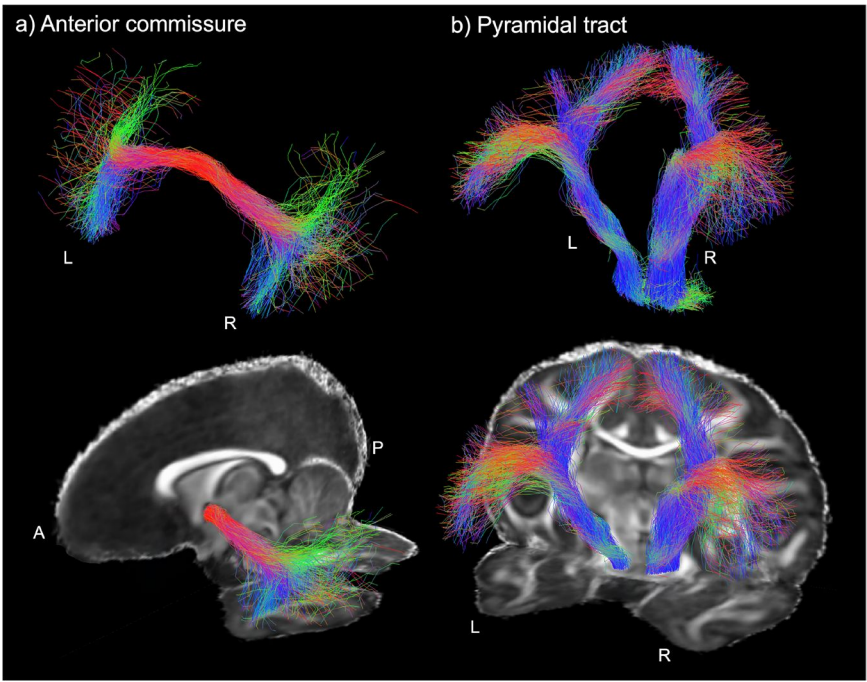


Fig. 6 Anterior commissure (a) and pyramidal tract (b, PyT) extracted from whole-brain tractograms and combined across all 9 subjects. Note that the PyT encompasses both the medial corticospinal tract (CST) and the lateral corticobulbar tract (CBT).

Animal ID	Sex	Age at anatomical MRI (years)	Age at diffusion MRI (years)	Weight (kg)
M01	M	23.0	22.5	11.5
M02	M	22.2	21.7	10.3
M03	F	13.5	13.7	4.0
M04	F	13.5	13.0	4.9
M05	M	4.8	4.6	8.5
M06	F	3.4	3.3	3.1
M07	F	3.4	4.1	2.8
M08	M	3.2	3.3	3.1
M09	M	3.0	3.1	2.5

Table 1. General information about the 9 macaque monkeys included in this dataset.

MR scanner, using the manufacturer’s body coil for transmission and 64-channel head coil for signal reception, as commonly used for human acquisitions. The animals were positioned in supine position, with their head maintained straight in the MRI coil using a shape memory head support that becomes rigid when activated with water (MOLDCARE head cushion, 20 × 45 cm, Alcare). Each animal underwent two MRI acquisition sessions dedicated to diffusion imaging (2 hours and 36 min) and anatomical imaging (1 hour and 11 minutes).

For each session, the animals were pre-medicated with intramuscular (IM) injections of diazepam (0.5 mg/kg) and atropine (0.05 mg/kg). For anatomical MRI sessions, they were anesthetized with ketamine (10 mg/kg, IM) and xylazine (1 mg/kg, IM) in the housing facility and were then transported to the imaging platform in a cart with a microwavable heating pad and blanket to maintain a stable body temperature. For diffusion MRI sessions, the animals were initially sedated with ketamine (10 mg/kg, IM), intubated, transported from the housing facility to the imaging platform, and were then maintained under anesthesia using 1–3% isoflurane mixed with pure oxygen. We used a regular veterinary anesthesia machine (VMS Plus, Midmark) placed in the MR control room, which was connected to the scanner room through 5-meter extension plastic tubing going through a penetration panel in the wall. Oxygen was delivered either from a dedicated supply circuit or from an oxygen concentrator (Nuvo Lite Mark 5, Nidek Medical). During all acquisitions, we continuously monitored heart rate and oxygen saturation using an MRI-compatible pulse oximeter (Model 7500, Nonin Medical) and respiratory rate using the manufacturer’s respiratory belt.

Diffusion-weighted imaging protocol. Diffusion-weighted acquisitions were obtained with an in-house MR pulse sequence designed to maximize SNR and provide high isotropic resolution while maintaining the total acquisition time below 3 hours for the wellness of the animal. The sequence was based on a 3D spin echo segmented EPI. The water-selective excitation pulse was a binomial pulse (1.3.3.1) with a bandwidth-time product (BWT) of 24. The refocusing pulse was a sinc pulse with a BWT of 3 and a slab thickness twice larger than the excitation pulse to ensure refocusing in the whole slab while minimizing specific absorption rate (SAR) and duration. The refocusing pulse was flanked with two crushers imposing a dephasing of 8π across one voxel to eliminate any transverse magnetization created by the refocusing pulse. Three navigators acquiring the centre line of the k-space with opposite direction for the second one were implemented between the excitation pulse and the refocusing pulse. They were used to correct for the N/2 Nyquist ghost and the phase variation across segments and partitions. Radiofrequency (RF) and gradient spoiling were implemented to enable shorter repetition times (TR). To minimize dropout and distortion artefacts, a partition was acquired in several TRs with interleaved segments. To further reduce the echo time (TE), a partial Fourier scheme was implemented such that the earlier part of the k-space was not acquired. Monopolar diffusion gradients of the same polarity were implemented on each side of the refocusing pulse.

The field of view was set to $100 \times 102.9 \times 61.6 \text{ mm}^3$ to cover the whole brain of all animals and was composed of 88 partitions. The nominal isotropic resolution was $0.7 \times 0.7 \times 0.7 \text{ mm}^3$. The repetition time was set to 700 ms to maintain a reasonable scan time. Consequently, the excitation flip angle was reduced to 70° maximizing SNR. By using 4 segments per partition and a partial Fourier factor of 7/8, the minimum TE of 64 ms could be used. The bandwidth per pixel was 776 Hz/pixel.

The protocol was repeated 32 times with a b-value of 1000 s/mm^2 and different diffusion encoding directions uniformly distributed across a sphere. Two additional scans were performed with the same time parameters but without diffusion weighting gradient. The phase encoding direction was inverted in the second one for subsequent distortion correction. In total, the acquisition lasted 2 hours and 36 minutes.

The sequence was made compatible with the standard vendors' reconstruction software to facilitate its use. The raw images were, therefore, available in real time.

Anatomical imaging protocol. In addition, high-resolution T1-weighted and T2-weighted acquisitions were collected in a separate session. The T1-weighted sequence was the vendor 3D MPRAGE sequence with the following key parameters: Sagittal orientation, field of view (FOV): $160 \times 160 \times 104 \text{ mm}^3$, phase oversampling of 4%, isotropic resolution of $0.5 \times 0.5 \times 0.5 \text{ mm}^3$, TE/TR = 2.32 ms / 2000 ms; 6 averages, inversion time of 1 s, non-selective inversion pulse, in-plane acceleration with GRAPPA (Generalized Autocalibrating Partially Parallel Acquisitions) factor of 2 including 24 lines of reference in the center, receive bandwidth of 250 Hz/pixel, train of 208 small flip angle (9°) water-selective excitations corresponding to the number of partitions. The total acquisition duration was 35 min and 50 sec.

The T2-weighted sequence was the vendor 3D SPACE sequence with the following key parameters: sagittal orientation, FOV: $160 \times 160 \times 104 \text{ mm}^3$, phase oversampling of 4%, isotropic resolution of $0.5 \times 0.5 \times 0.5 \text{ mm}^3$, TE/TR = 384 ms / 3200 ms, 4 averages, in-plane acceleration with GRAPPA factor of 2 with 24 lines of reference, fat saturation preparation module, receive bandwidth of 473 Hz/pixel, non-selective refocusing pulse train of 218 pulses with variable flip angles. This sequence also had an acquisition duration of 36 min and 19 sec.

Data processing. All images were processed using *versaFlow*²⁵ (<https://github.com/AlexVCaron/versaFlow>), a pipeline designed for the preprocessing and analysis of diffusion MRI data from macaque brains. Specific configuration parameters used to produce the MADI@700 dataset are available in the Supplementary Materials (*nex-tflow.config*).

Brain mask generation. Brain masks were extracted from the T1-weighted images, which were first imported into ITK-SNAP (<http://www.itksnap.org/pmwiki/pmwiki.php>) and converted into a nifti file. Since the MRI acquisition contains Rician noise, the nifti file was denoised using the MATLAB Toolbox "MRIDenoisingPackage" with an Adaptive Optimized Nonlocal Means (AONLM) filter⁴⁰ (<https://sites.google.com/site/pierrickcoupe/software/denoising/mri-denoising/mri-denoising-software?authuser=0>). The denoised images were then used as an input to the generalizable brain extraction net (BEN) algorithm⁴¹, a recent deep neural network for extracting brain tissues from multimodal MRI images across species (<https://github.com/yu02019/BEN>). The obtained masks were then verified on the corresponding MRI volumes. Some masks obtained upon initial application of the BEN algorithm were suboptimal, which led us to manually refine them, retrain the BEN weights, and generate them again using the updated model⁴². The final masks were finally imported into ITK-SNAP and smoothened with a standard deviation of $1 \times 1 \times 1 \text{ mm}^3$ while selecting the 'smooth all frames' option to remove small missing pixels. Since T1-weighted and T2-weighted images were acquired in the same session, we used the same mask on both image sets.

T1- and T2-weighted image preprocessing. A short preprocessing sequence was used to filter background noise and intensity bias in T1-weighted images and to perform tissue segmentation into white matter (WM), grey matter (GM), and cerebrospinal fluid (CSF). Additionally, registration utilities from ANTs (Advanced Normalization Tools)⁴³ were employed to project various brain atlases (INIA19³⁵, D99³⁶, SARM³⁷, CHARM³⁸) into the individual subject space.

Diffusion-weighted image preprocessing. Diffusion MRI preprocessing included filtering for background noise, magnetic susceptibility, eddy currents, motion, and intensity bias. Diffusion profiles were obtained by Diffusion

Tensor Imaging (DTI)⁴⁴, yielding a Diffusion Tensor (DT), and by Constrained Spherical Deconvolution (CSD)⁴⁵, leading to a fiber Orientation Distribution Function (fODF). The diffusion profiles (DT and fODF) were then used to compute various diffusion metrics: mean, axial and radial diffusivities (DT-MD, DT-AD, DT-RD), fractional anisotropy (DT-FA), RGB maps of the direction of maximum diffusivity (DT-RGB and fODF-RGB), apparent fiber density (fODF-AFD), number of fiber orientations (fODF-NuFO), and fODF peaks.

Tractography. Probabilistic local tractography^{22,46} was performed with a GPU implementation available in the Sherbrooke Connectivity Imaging Lab (SCIL) Python dMRI processing toolbox (Scilpy) (<https://github.com/scilus/scilpy>) on the whole brain. To optimize coverage and resolution, streamlines were seeded at 5 points in each valid voxel and propagated with a step of 100 μm and 20 degrees of maximal angular difference⁴⁷. Tracking positions were discretized using the “repulsion724” sphere available in the Diffusion Imaging in Python (Dipy) library⁴⁸. Streamlines that extended more than 100 mm were rejected as they would reach outside the brain volume. Valid tracking execution and seed positions were restricted to voxels within the WM mask where the DT-FA was higher than 0.2 and the DT-RD lower than 0.0017 mm^2/s . Inconsistencies in WM coverage from the segmentation obtained in *versaFlow* (e.g., WM mask that sometimes extended into the interhemispheric space or missed coverage in the brainstem, cerebellum, and around deep grey matter structures) caused false positives and/or negatives in some subjects. To circumvent this issue, the tracking and seeding masks were fixed using atlases registered by *versaFlow*. Specifically, the brainstem, cerebellum, and deep grey matter regions were identified using all labels from the SARM atlas³⁷. The interhemispheric space was eliminated from the mask by removing the veins - label #6 - and CSF - label #1 - of the CHARM atlas³⁸.

Common template space creation. The common template space was generated from all 9 subjects using a multi-modal registration approach available in ANTs⁴³. Modalities and metrics used to generate the template included: anatomical T1 and T2 images, $b = 0 \text{ s/mm}^2$ and powder-averaged⁴⁹ $b = 1000 \text{ s/mm}^2$ shells, brain masks, DT-AD, DT-RD, DT-FA, fODF-AFD in the WM, fODF-AFD in total, as well as the INIA19 and D99 atlases registered in subject space (note that the INIA19 and D99 atlases are both made available as outputs of the *versaFlow* pipeline, which uses the AFNI NMT v2.0 template to register them to subject space). Using such multi-modal targets makes the resulting template more precise and versatile when performing registration operations between subject and template space, allowing it to compute the transformations using only diffusion or anatomical information or the combination of both. Once computed, DT-MD and DT-RGB were added to the template by registration and majority voting. Additionally, the Mean Squared Error (MSE) and variance between the generated template and the subject's spaces are computed for each modality provided.

Bundle extraction in the common template space. Each of the 9 whole-brain tractograms was registered in the template space using both linear and non-linear deformation matrices obtained during the creation of the template space. The parcellation of the INIA19 atlas³⁵ was used as regions of interest (ROIs) to extract some bundles of interest, namely the corpus callosum (CC) (Fig. 5), the anterior commissure (AC), and the pyramidal tract (PyT) (Fig. 6). Additional ROIs were built by combining INIA19 ROIs. All steps applied to extract these different bundles utilized the Scilpy toolbox (<https://github.com/scilus/scilpy>). In each bundle extraction, all streamlines with a maximum looping angle superior to 270° were systematically removed.

Corpus callosum (CC). We first extracted the streamlines passing through the callosal ROIs but not through the basal ganglia, thalamus, fornix, and ventricles. The subsequent callosal streamlines were extracted for each lobe (frontal, parietal, occipital, and temporal), segmenting the CC into four homotopic lobar sub-tracts. For example, the homotopic frontal CC encompassed the streamlines with one termination in the left frontal lobe and the other in the right one.

Anterior commissure (AC). All the streamlines passing through the left and right ROIs of the anterior commissure and terminating either in the left or in the right temporal lobe were first extracted, while streamlines passing through the extreme capsule and the frontal white matter ROIs were removed. Only streamlines with a maximum length of 70 mm were finally considered as constituting the AC pathway.

Pyramidal tract (PyT). We obtained the main part of the PyT encompassing the precentral gyrus⁵⁰ by extracting the streamlines with one termination in the precentral gyrus and passing ipsilaterally through the internal capsule and the cerebral peduncle. We excluded the streamlines passing through the postcentral and superior frontal gyri, as well as the cerebellum. These steps were applied separately for each hemisphere.

Data Records

The dataset is openly accessible through the EBRAINS infrastructure (<https://doi.org/10.25493/XGXR-BRD>)³⁹, as well as the PRIME-DE database (https://fcon_1000.projects.nitrc.org/indi/PRIME/MADI@700.html). The entire dataset occupies 12 GB of disk space. The dataset is compliant with the Brain Imaging Data Structure (BIDS). Its organization is described in Fig. 2. The root of the dataset is divided into 10 directories corresponding to 9 animals (M01 to M09) and the derivatives. The derivatives directory contains the processing of the 9 animals by *versaFlow* and the common space template. All 3D and 4D data are stored in the archived nifti format (*.nii.gz). Diffusion directions and weighting are stored following the FSL convention (.bval/.bvec). Transformation matrices between the subject space and the common space template and back are provided alongside the template. Tractograms are stored in trk format.

Animal ID	Mean absolute translation			Mean absolute rotation			T1w CNR	b0 SNR
	x	y	z	x	y	z		
	[mm]	[mm]	[mm]	[°]	[°]	[°]		
M01	0,0093	0,0225	0,1164	3,90E-4	5,60E-4	3,90E-4	0,6724	4,3254
M02	0,0034	0,0179	0,0662	3,26E-4	2,75E-4	3,26E-4	0,5828	4,4523
M03	0,0223	0,0913	0,0493	8,17E-5	1,62E-4	8,17E-5	0,6528	4,2043
M04	0,0052	0,1746	0,0744	2,62E-4	5,03E-4	2,62E-4	1,0953	4,1992
M05	0,0056	0,0804	0,0184	3,42E-4	1,88E-4	3,42E-4	0,8762	4,3637
M06	0,0084	0,1078	0,0480	2,86E-4	1,32E-4	2,86E-4	0,7276	3,8055
M07	0,0115	0,1050	0,1074	1,15E-4	3,08E-4	1,15E-4	0,6676	4,2409
M08	0,0056	0,1301	0,0601	2,52E-4	2,03E-4	2,52E-4	0,6466	4,6500
M09	0,0110	0,1123	0,0478	3,18E-4	2,41E-4	3,18E-4	0,6641	3,3488
Mean	0,0093	0,0225	0,1164	3,90E-4	5,60E-4	3,90E-4	0,7317	4,1767
Median	0,0034	0,0179	0,0662	3,26E-4	2,75E-4	3,26E-4	0,6676	4,2409
Max	0,0223	0,0913	0,0493	8,17E-5	1,62E-4	8,17E-5	1,0953	4,6500

Table 2. Quantitative measurement of image quality. Mean absolute translation and rotation are reported by Eddy. CNR in the anatomical T1w image and SNR in the b0 image are computed using the MRIQC Python library⁵⁷.

Technical Validation

Qualitative control assessment was performed for the 9 datasets at each step of the *versaFlow* pipeline, using the Diffusion MRI Quality Check in the Python (dmriQcPy) toolbox (<https://github.com/scilus/dmriqcPy>)⁵¹. These checks ensure proper motion correction, the absence of outliers, and the precise alignment between diffusion space, anatomical space, and the template. The results of the quality assessment are available on Zenodo⁵². In addition, quantitative estimates of motion and image quality are reported in Table 2 to support quality of the proposed dataset. The rigid transformation estimated with FSL Eddy to coregister the diffusion weighted acquisitions was extracted to assess the degree of motion. The averages of the absolute values of translation and rotation in each axis are provided for each animal. The contrast-to-noise ratio (CNR) was computed on the anatomical T1 weighted acquisitions to assess the contrast between white and grey matter (T1wCNR). The SNR was computed within the white matter in the EPI acquisitions with $b = 0 \text{ s/mm}^2$ (b0SNR). No correlation between the age or the weight and the quality control metrics was found ($R < 0.55$) suggesting that the quality of the image is not driven by the age or the weight of the subject in this dataset.

To ascertain the quality of the template built from the 9 subjects in this study, a variance map was computed for each modality included, by registering all subjects' images to the template and computing the variance in each voxel (all variance maps can be found in the following folder of the dataset: derivatives/madi@700-template/). The resulting map was then inspected visually to ensure the complete convergence of the template.

Technical validation of MADI@700 also occurred at the final stage of our pipeline, *i.e.*, the reconstruction of whole-brain tractograms and the extraction of specific fiber bundles (CC, AC, and PyT). Indeed, we observed that the bundles extracted from our dataset follow the known macaque monkey neuroanatomy:

- Figure 5 (top) shows the callosal streamlines extracted from the whole-brain tractograms registered in the template space for the 9 subjects. The color-coded concatenation of these callosal streamlines is displayed in the bottom row and corresponds to the frontal (red), parietal (green), occipital (blue), and temporal (yellow) topological organization of the homotopic callosal fibers as previously described⁵³.
- Figure 6a shows the AC extracted for the 9 subjects. As previously described, it traverses the midline as a compact and prominent fiber bundle connecting the first third of the temporal cortex⁵³.
- Figure 6b shows the pyramidal tract across the 9 subjects, focusing on its main section covering the precentral gyrus. It corresponds well to the cortico-descending projections previously described⁵⁰. Notably, we observed a thorough fanning along the precentral gyrus, encompassing both the medial corticospinal tract (CST), which monitors limb and trunk movements, and the lateral corticobulbar tract (CBT), which controls movements of the face, head, and neck. Note that the current template-based PyT may be improved in a bundle-specific tractography approach⁵⁴, as previously applied in the human brain to create a population-based atlas of the human pyramidal tract⁵⁵.

Finally, we report in Supplementary Table 1 comprehensive quantitative metrics characterizing various structural attributes of the different white matter fiber bundles presented here as examples (CC, AC, and PyT). For each of these fiber bundles, data from individual subjects (M01-M09) are summarized, including streamline count, tract volume, average streamline length, minimum and maximum streamline lengths, span, curl, and elongation, calculated through a shape analysis methodology⁵⁶.

These metrics characterize different structural aspects of fiber bundles: “length” measures the average pathway trajectory; “volume” quantifies the space occupied by the fiber bundle; “span” describes the straight-line distance between bundle endpoints; “curl”, computed as the ratio length/span, assesses the curvature of the fiber

trajectories (a higher curl indicates more curved fibers); and “elongation”, computed as the ratio length/diameter reflects how elongated the fiber bundle is.

For the CC, data are presented separately for the frontal, parietal, occipital, and temporal lobes. The frontal CC displays the highest mean streamline count (2050.1 ± 855.8) and volume ($3656.5 \pm 1026.0 \text{ mm}^3$), with moderate average streamline length ($42.3 \pm 2.5 \text{ mm}$) and curl (0.4 ± 0.0), suggesting relatively direct but voluminous connections. The parietal CC has moderate streamline count (559.9 ± 196.0) and volume ($1439.8 \pm 407.8 \text{ mm}^3$), with higher curl (0.5 ± 0.0) and elongation values (7.3 ± 1.1), reflecting moderately elongated and curved fibers. Similarly, the occipital CC shows moderate streamline count (505.3 ± 326.8) and volume ($1008.7 \pm 425.1 \text{ mm}^3$), with notable curl (0.5 ± 0.0) and the highest average elongation (10.0 ± 2.5) among CC subdivisions, indicating highly elongated and curved trajectories. Conversely, the temporal CC shows the longest average streamline length ($61.0 \pm 5.0 \text{ mm}$), highest elongation (30.0 ± 19.2), and lowest streamline count (100.0 ± 108.2), indicative of fewer but highly elongated fiber trajectories.

As expected with its anatomical definition, the AC exhibits relatively elongated streamlines (elongation 33.5 ± 26.4) and moderately high curl values (0.4 ± 0.0), reflecting significant curvature and elongation in fibers connecting the temporal lobes across hemispheres. Mean streamline count and volume were lower than frontal CC but greater than temporal CC, suggesting intermediate connectivity density.

Data for the PyT are provided separately for the left and right hemispheres. Both sides demonstrate similar structural characteristics, with comparable average streamline length (approximately 48 mm) and moderate elongation (left: 17.2 ± 3.2 , right: 18.6 ± 6.6), indicating relatively direct yet somewhat elongated trajectories typical of major descending motor pathways.

No strong correlation ($R < 0.8$) was observed between those tractography metrics and the b0SNR except in the AC where the correlation with the curl parameter reached 0.85. The overall low correlation factor suggests that the tractography metrics are not biased by the SNR of the data.

Collectively, these metrics facilitate detailed morphological analyses and comparisons of white matter connectivity patterns across subjects or brain regions, providing crucial insights into structural connectivity and its implications for brain function.

Usage Notes

The dataset can be used to investigate anatomical features of macaque brains. This can be particularly helpful in preparation for surgery or neurostimulation. The dataset can also be used to validate any methods aiming to characterize the structures of the brain. A standard nifti viewer can be used to visualize the images, and the Dipy and Scilpy (<https://github.com/scilus/scilpy>) toolboxes are recommended for analyzing the tractograms.

Code availability

The processing pipeline used to generate the output data is available at <https://github.com/AlexVCaron/versaFlow> and only requires open-source packages to be ran (ANTs, etc.). The output data were obtained with release 1.0.0 of the code. The compiled MR pulse sequence is freely available on the Siemens Teamplay C2P exchange platform (DW-3D-SE-EPI sequence, VE11C version).

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Author contributions

L.P. and F.B.W. designed and coordinated the study, in collaboration with M.D. and S.M. N.C. designed the 3D EPI sequence used for all diffusion MRI acquisitions. N.C., A.B., T.H.N., S.A., S.M., L.P., F.B.W. prepared the animals and acquired the data. A.V.C. and M.D. developed the data processing pipeline versaFlow for processing

of macaque diffusion MRI data. A.V.C., N.C., A.G., L.P. analyzed the data. A.V.C., N.C., L.P., F.B.W. wrote the first version of the manuscript, which was then edited and approved by all authors.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to L.P. or F.B.W.

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Supplementary table 1. Individual and Mean ($\pm$ standard deviation, S.D.) bundle descriptors values for the four lobar parts of the corpus callosum (CC), anterior commissure (AC), and pyramidal tract (PyT). Lengths and span measures are reported in millimeters (mm), volume in cubic millimeters (mm³), and curl and elongation are expressed as ratios—length/span and length/diameter, respectively (Yeh, 2020).

Frontal CC

	Streamlines count	Volume	Average length	Min length	Max length	Span	Curl	Elongation
M01	1886	3434.3	43.4	20.3	90.0	105.1	0.41	4.3
M02	1810	3316.8	42.3	21.0	83.9	99.1	0.43	4.2
M03	1306	2531.0	37.6	15.8	87.3	75.8	0.50	4.1
M04	2528	4542.4	43.0	17.2	86.1	100.2	0.43	3.7
M05	2633	4342.1	43.7	17.6	94.2	99.8	0.44	3.9
M06	3429	4842.2	40.2	14.0	92.0	100.8	0.40	3.2
M07	1107	2794.4	45.9	14.2	95.8	107.1	0.43	5.2
M08	2825	4887.7	44.0	11.8	89.5	103.6	0.42	3.7
M09	927	2217.8	40.5	18.4	72.1	78.4	0.52	4.9
Mean	2050.1	3656.5	42.3	16.7	87.9	96.7	0.4	4.1
S.D.	855.8	1026.0	2.5	3.0	7.0	11.4	0.0	0.6

Parietal CC

	Streamlines count	Volume	Average length	Min length	Max length	Span	Curl	Elongation
M01	716	1907.8	48.6	17.5	86.7	109.1	0.45	6.9
M02	295	856.9	43.7	16.3	69.5	86.1	0.51	8.8
M03	566	1249.7	40.6	18.8	69.1	82.8	0.49	6.5
M04	846	2007.8	50.3	22.9	96.7	119.0	0.42	7.1
M05	736	1641.4	42.6	16.1	66.7	84.3	0.50	6.1
M06	559	1626.3	47.6	23.4	73.6	106.3	0.45	7.2
M07	331	1096.1	49.2	24.2	80.4	101.1	0.49	9.2
M08	635	1579.3	42.4	13.5	86.8	83.9	0.51	6.2
M09	355	992.7	43.1	17.0	69.9	79.3	0.54	8.0
Mean	559.9	1439.8	45.4	18.9	77.7	94.7	0.5	7.3
S.D.	196.0	407.8	3.6	3.8	10.4	14.3	0.0	1.1

Occipital CC

	Streamlines count	Volume	Average length	Min length	Max length	Span	Curl	Elongation
M01	829	1496.9	48.5	20.0	87.3	96.9	0.50	7.7
M02	368	860.7	49.9	20.8	84.6	110.1	0.45	10.6
M03	138	435.0	49.2	24.4	73.9	94.8	0.52	14.6
M04	1105	1740.6	47.3	13.8	86.5	99.5	0.48	6.9
M05	367	853.1	48.8	15.0	78.9	104.0	0.47	10.4
M06	196	570.5	46.5	23.2	71.2	94.7	0.49	11.7
M07	799	1300.2	44.8	12.9	86.3	95.2	0.47	7.4
M08	370	918.0	53.8	24.6	86.4	109.2	0.49	11.6
M09	376	903.1	46.8	15.9	74.3	92.7	0.51	9.4
Mean	505.3	1008.7	48.4	19.0	81.0	99.7	0.5	10.0
S.D.	326.8	425.1	2.6	4.6	6.5	6.6	0.0	2.5

Temporal CC

	Streamlines count	Volume	Average length	Min length	Max length	Span	Curl	Elongation
M01	235	871.5	54.2	35.8	85.9	165.7	0.33	12.0
M02	18	96.6	58.6	33.8	85.4	140.8	0.42	40.4
M03	29	147.8	58.1	43.5	83.0	148.4	0.39	32.2
M04	308	1244.6	64.6	30.0	103.3	178.2	0.36	13.0
M05	8	52.8	71.9	50.4	94.5	153.2	0.47	74.3
M06	70	350.8	61.3	30.7	93.2	165.7	0.37	22.7
M07	42	212.8	58.3	35.8	92.0	167.5	0.35	27.0
M08	157	640.5	60.8	31.1	96.6	158.5	0.38	16.6
M09	33	176.9	61.1	44.5	86.5	159.5	0.38	31.8
Mean	100.0	421.6	61.0	37.3	91.2	159.7	0.4	30.0
S.D.	108.2	411.2	5.0	7.2	6.5	11.2	0.0	19.2

AC

	Streamlines count	Volume	Average length	Min length	Max length	Span	Curl	Elongation
M01	32	184.0	70.9	51.1	100.1	158.8	0.45	39.0
M02	106	518.0	73.1	45.4	100.2	159.5	0.46	24.3
M03	42	190.5	61.9	46.1	94.7	133.0	0.47	31.3
M04	86	427.4	74.0	45.4	106.3	167.9	0.44	27.3
M05	203	792.6	72.4	44.5	101.2	166.8	0.43	19.4
M06	171	642.4	65.2	41.8	94.6	144.6	0.45	18.4
M07	157	706.4	72.0	47.6	100.6	166.0	0.43	20.4
M08	126	481.7	63.2	42.9	99.0	146.6	0.43	20.3
M09	5	35.2	77.3	62.4	90.0	170.0	0.45	101.6
Mean	103.1	442.0	70.0	47.5	98.5	157.0	0.4	33.5
S.D.	67.8	258.8	5.3	6.2	4.8	12.8	0.0	26.4

Left PyT

	Streamlines count	Volume	Average length	Min length	Max length	Span	Curl	Elongation
M01	50	192.6	49.6	36.4	65.6	151.4	0.33	22.3
M02	211	582.3	50.4	30.2	68.5	161.5	0.31	13.2
M03	54	192.7	44.9	30.4	57.6	134.1	0.34	19.2
M04	92	326.0	51.5	31.6	66.6	154.4	0.33	18.1
M05	95	319.0	45.5	31.8	67.4	143.2	0.32	15.2
M06	135	423.8	46.7	29.6	68.0	130.7	0.36	13.7
M07	155	470.2	49.9	33.4	64.6	157.0	0.32	14.4
M08	49	171.3	44.9	31.6	57.7	131.5	0.34	20.4
M09	72	260.6	47.9	36.0	58.6	126.9	0.38	18.2
Mean	101.4	326.5	47.9	32.3	63.8	143.4	0.3	17.2
S.D.	55.5	141.4	2.5	2.5	4.6	13.0	0.0	3.2

Right PyT

	Streamlines count	Volume	Average length	Min length	Max length	Span	Curl	Elongation
M01	26	112.2	50.2	32.7	73.1	145.5	0.35	29.8
M02	223	613.9	48.4	27.9	68.5	146.9	0.33	12.1
M03	52	176.2	43.7	29.2	61.2	132.4	0.33	19.3
M04	31	137.0	52.8	41.0	68.4	144.5	0.37	29.0
M05	118	428.0	50.7	30.8	67.6	146.3	0.35	15.5
M06	175	469.7	45.0	35.5	57.1	131.1	0.34	12.4
M07	132	438.0	50.8	38.1	77.5	154.4	0.33	15.3
M08	71	240.3	47.5	39.7	63.0	129.0	0.37	18.7
M09	82	268.7	43.6	27.7	58.8	117.9	0.37	15.5
Mean	101.1	320.4	48.1	33.6	66.1	138.7	0.3	18.6
S.D.	66.9	173.5	3.4	5.1	6.7	11.6	0.0	6.6